A REPAIRED-GAUGE COMPACTNESS CRITERION FOR TYPE II RENORMALIZATION IN 3D NAVIER–STOKES

Abstract. We prove a compact Type II renormalization criterion for the three-dimensional incompressible Navier–Stokes equations. The theorem is formulated for renormalized orbits satisfying explicit compactness, pressure, modulation, and local energy hypotheses. Under these hypotheses, finite total renormalization cost is impossible. The proof combines the renormalized local energy identity, a Caccioppoli estimate, pressure estimates modulo constants, local time regularity, Aubin–Lions–Simon compactness, a repaired weighted-moment scale gauge, and a compact low-dissipation rigidity argument.

1. Introduction

This paper proves a compact renormalization criterion for the three-dimensional incompressible Navier–Stokes equations. The analysis is carried out in renormalized variables associated with a concentrating scale and center. Under compactness, pressure, modulation, and local energy hypotheses stated below, the localized renormalization cost has infinite total accumulation. The proof is based on the renormalized local energy identity, pressure estimates modulo constants, compactness in local critical topologies, and a rigidity mechanism for compact low-dissipation subsequences.

1.1. Strategy overview.

The proof strategy is:

(1) Renormalize the Navier–Stokes flow around a concentrating candidate singularity:

$$u(x,t) = \lambda(t)^{-1} V\left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t)\right), \qquad \tau_t = \lambda(t)^{-2}.$$

(2) In renormalized variables, define the nonnegative renormalization cost

$$\tilde{\mathfrak{D}}_{R_0}(\tau) := \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\, dy,$$

where $\phi_{R_0} \geq 0$ is a cutoff satisfying $\phi_{R_0} \equiv 1$ on B_{R_0}, and $a_+(\tau) = \max(a(\tau), 0)$.

(3) Prove that if the total cost were finite, then by nonnegativity there exists a sequence $\tau_n \to \infty$ such that

$$\tilde{\mathfrak{D}}_{R_0}(\tau_n) \to 0.$$

(4) Show that low cost implies low localized dissipation:

$$\tilde{\mathfrak{D}}_{R_0}(\tau_n) \to 0 \quad \Longrightarrow \quad \int_{B_{R_0}} |\nabla V(\tau_n)|^2 \to 0.$$

(5) Use compactness + time regularity (Aubin–Lions–Simon) to extract a strong local limit of $V(\tau_n)$.

(6) Prove the rigidity theorem: a compact normalized L^3-tight orbit cannot admit a subsequence with vanishing localized dissipation on a fixed core.

(7) Conclude that finite total renormalization cost is impossible:

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

The hypotheses entering the final theorem are stated explicitly in the renormalized variables.

1.2. **Theorem map.**

1.2.1. *Main theorem.* **Theorem A.** Under hypotheses (H1)–(H5) in the main theorem below,

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

1.2.2. *Internal lemmas used.*

(1) Lemma 1: renormalized equation.
(2) Lemma 2: pointwise renormalized local energy identity.
(3) Lemma 3: spacetime cutoff renormalized local energy identity.
(4) Lemma 4: localized renormalized energy identity with spatial cutoff.
(5) Lemma 5: renormalized Caccioppoli estimate.
(6) Lemma 6: nonnegative integrable cost has a vanishing subsequence.
(7) Lemma 7: low renormalization cost implies low core dissipation.
(8) Lemma 8: compact normalized low-dissipation subsequence is impossible.
(9) Lemma 9: local $L^{3/2}$ pressure estimate modulo a constant.
(10) Lemma 9.1: uniform pressure-tail control from global L^3.
(11) Lemma 9.2: verification of the local L^2 pressure hypotheses.
(12) Lemma 9.3: local L^2 pressure estimate modulo constants.
(13) Lemma 10: local $L^2_\tau H^{-1}_x$ time-regularity.
(14) Lemma 11: Aubin–Lions–Simon closure and contradiction.

1.2.3. *Gauge appendix lemmas.* G1. Repaired weighted-moment scale gauge and centering gauge. G2. Fréchet derivative of G_0 and exact scaling derivative. G3. Modulation matrix entry formulas. G4. Continuity of the modulation matrix (including weighted-moment entries). G5. Full matrix invertibility by Schur complement under quantitative hypotheses. G6. Bounded modulation parameters under uniform nondegeneracy.

1.3. **Standard external tools quoted.** The following standard results are used as black boxes:

(1) **Calderón–Zygmund / Riesz transform boundedness**

$$\|\mathcal{R}_i\mathcal{R}_j f\|_{L^p(\mathbb{R}^3)} \le C_p\|f\|_{L^p(\mathbb{R}^3)}, \qquad 1 < p < \infty.$$

(2) **Sobolev and bounded-domain interpolation**

$$H^1(B_R) \hookrightarrow L^6(B_R),$$

and

$$\|u\|_{L^4(B_R)} \le C(R)\|u\|^{1/4}_{L^2(B_R)}\|u\|^{3/4}_{L^6(B_R)}.$$

(3) **Aubin–Lions–Simon compactness theorem**.
(4) **Lions–Magenes trace regularity**

$$L^2(0,1;H^1(B_R)) \cap H^1(0,1;H^{-1}(B_R)) \subset C([0,1];L^2(B_R)).$$

We state precisely where each is used.

2. Renormalization

2.1. **Lemma 1 (Renormalized equation).** Let u, p solve

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p = \nu \Delta_x u, \qquad \nabla_x \cdot u = 0,$$

and define

$$u(x,t) = \frac{1}{\lambda(t)} V(y,\tau), \qquad y = \frac{x - x_c(t)}{\lambda(t)}, \qquad \frac{d\tau}{dt} = \frac{1}{\lambda(t)^2},$$

with

$$p(x,t) = \frac{1}{\lambda(t)^2} P(y,\tau).$$

Assume u, p are classical, $x_c, \lambda \in C^1$, and $\lambda > 0$. Then

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)\big(V + y \cdot \nabla V\big) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0,$$

where

$$a(\tau) = \lambda(t)\lambda'(t), \qquad b(\tau) = \lambda(t) x_c'(t).$$

2.1.1. *Proof.* We compute

$$y = \frac{x - x_c(t)}{\lambda(t)} \quad \Longrightarrow \quad \partial_t y = -\frac{x_c'(t)}{\lambda(t)} - \frac{\lambda'(t)}{\lambda(t)} y.$$

Also,

$$u(x,t) = \lambda^{-1} V(y,\tau),$$

so

$$\partial_t u = -\frac{\lambda'}{\lambda^2} V + \frac{1}{\lambda}\Big(\partial_\tau V\, \tau_t + (\partial_t y) \cdot \nabla V\Big).$$

Since $\tau_t = \lambda^{-2}$,

$$\partial_t u = \frac{1}{\lambda^3} \partial_\tau V - \frac{1}{\lambda^2} x_c'(t) \cdot \nabla V - \frac{\lambda'}{\lambda^2}(V + y \cdot \nabla V).$$

Also,

$$\nabla_x u = \frac{1}{\lambda^2} \nabla_y V, \qquad \Delta_x u = \frac{1}{\lambda^3} \Delta_y V, \qquad (u \cdot \nabla_x)u = \frac{1}{\lambda^3}(V \cdot \nabla_y)V,$$

and

$$\nabla_x p = \frac{1}{\lambda^3} \nabla_y P.$$

Substituting into

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p = \nu \Delta_x u$$

and multiplying by λ^3 gives

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + \lambda\lambda'(V + y \cdot \nabla V) + \lambda x_c'(t) \cdot \nabla V.$$

Thus

$$a(\tau) = \lambda\lambda'(t), \qquad b(\tau) = \lambda x_c'(t),$$

and $\nabla \cdot V = 0$ follows from $\nabla_x \cdot u = 0$.
□

3. Renormalized local energy identities

Set

$$e(y,\tau) := \frac{1}{2}|V(y,\tau)|^2.$$

3.1. **Lemma 2 (Pointwise renormalized local energy identity).** Assume V, P are classical solutions of the renormalized equation. Then

$$\partial_\tau e - \nu\Delta e + \nu|\nabla V|^2 + \operatorname{div}\big((e+P)V - a\,ye - b\,e\big) + ae = 0.$$

3.1.1. *Proof.* Take the dot product of

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(V + y\cdot\nabla V) + b\cdot\nabla V$$

with V. We use:

$$V\cdot\partial_\tau V = \partial_\tau e, \qquad V\cdot(V\cdot\nabla V) = V\cdot\nabla e = \operatorname{div}(eV),$$

$$V\cdot\nabla P = \operatorname{div}(PV) \quad (\nabla\cdot V = 0),$$

$$\nu V\cdot\Delta V = \nu\Delta e - \nu|\nabla V|^2,$$

$$V\cdot(V + y\cdot\nabla V) = 2e + y\cdot\nabla e,$$

$$V\cdot(b\cdot\nabla V) = b\cdot\nabla e = \operatorname{div}(be),$$

since $b = b(\tau)$ depends only on τ. Therefore

$$\partial_\tau e + \operatorname{div}(eV) + \operatorname{div}(PV) = \nu\Delta e - \nu|\nabla V|^2 + a(2e + y\cdot\nabla e) + \operatorname{div}(be).$$

Now

$$2e + y\cdot\nabla e = \operatorname{div}(ye) - e,$$

because $\operatorname{div}(ye) = 3e + y\cdot\nabla e$. Substituting gives the claim:

$$\partial_\tau e - \nu\Delta e + \nu|\nabla V|^2 + \operatorname{div}\big((e+P)V - a\,ye - be\big) + ae = 0.$$

□

3.2. **Lemma 3 (Spacetime cutoff renormalized local energy identity).** Let $\zeta = \zeta(y,\tau) \in C_c^\infty(\mathbb{R}^3\times(\tau_1,\tau_2))$, $\zeta \geq 0$. Under the assumptions of Lemma 2,

$$\begin{aligned}
&\int_{\mathbb{R}^3} e(y,\tau_2)\zeta(y,\tau_2)\,dy - \int_{\mathbb{R}^3} e(y,\tau_1)\zeta(y,\tau_1)\,dy \\
&\quad + \nu\int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} |\nabla V|^2\zeta\,dy\,d\tau + \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} a(\tau)e\zeta\,dy\,d\tau \\
&= \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} e(\partial_\tau\zeta + \nu\Delta\zeta)\,dy\,d\tau + \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} (e+P)V\cdot\nabla\zeta\,dy\,d\tau \\
&\quad - \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} a(\tau)e\,y\cdot\nabla\zeta\,dy\,d\tau - \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} e\,b(\tau)\cdot\nabla\zeta\,dy\,d\tau.
\end{aligned}$$

3.2.1. *Proof.* Multiply the identity in Lemma 2 by ζ, integrate over $\mathbb{R}^3\times(\tau_1,\tau_2)$, and integrate by parts in both τ and y. Since ζ is compactly supported in $(\tau_1,\tau_2)\times\mathbb{R}^3$, no boundary terms arise except those displayed at $\tau = \tau_1, \tau_2$. The formula follows directly.
□

3.3. **Lemma 4 (Localized renormalized energy identity).** Let $\phi \in C_c^\infty(\mathbb{R}^3)$, $0 \le \phi \le 1$, and define

$$\mathcal{E}_\phi(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi(y)\, dy.$$

Assume:

- $V(\cdot,\tau) \in H^2(\mathbb{R}^3) \cap L^3(\mathbb{R}^3)$,
- $P(\cdot,\tau) \in H^1_{\mathrm{loc}}(\mathbb{R}^3)$,
- V is continuous in τ with values in $H^1 \cap L^3$,
- a, b are continuous.

Then $\mathcal{E}_\phi \in C^1$ and

$$\begin{aligned}\frac{d}{d\tau}\mathcal{E}_\phi(\tau) = &-\nu \int |\nabla V|^2 \phi + \frac{\nu}{2} \int |V|^2 \Delta\phi + \frac{1}{2} \int |V|^2 V \cdot \nabla\phi + \int P\, V \cdot \nabla\phi \\ &- \frac{a}{2} \int |V|^2 \phi - \frac{a}{2} \int |V|^2 y \cdot \nabla\phi - \frac{1}{2}\, b \cdot \int |V|^2 \nabla\phi.\end{aligned}$$

3.3.1. *Proof.* Differentiate:

$$\frac{d}{d\tau}\mathcal{E}_\phi = \int V \cdot \partial_\tau V\, \phi.$$

Insert the renormalized equation:

$$\frac{d}{d\tau}\mathcal{E}_\phi = \nu \int V \cdot \Delta V\, \phi - \int V \cdot (V \cdot \nabla V)\phi - \int V \cdot \nabla P\, \phi + a \int V \cdot (V + y \cdot \nabla V)\phi + b \cdot \int V \cdot \nabla V\, \phi.$$

Compute each term.
For the viscous term,

$$\int V \cdot \Delta V\, \phi = -\int |\nabla V|^2 \phi + \frac{1}{2} \int |V|^2 \Delta\phi.$$

For convection, using $\nabla \cdot V = 0$,

$$-\int V \cdot (V \cdot \nabla V)\phi = \frac{1}{2} \int |V|^2 V \cdot \nabla\phi.$$

For pressure,

$$-\int V \cdot \nabla P\, \phi = \int P\, V \cdot \nabla\phi.$$

For the scale term,

$$\int V \cdot (V + y \cdot \nabla V)\phi = -\frac{1}{2} \int |V|^2 \phi - \frac{1}{2} \int |V|^2\, y \cdot \nabla\phi.$$

For the translation term,

$$b \cdot \int V \cdot \nabla V\, \phi = -\frac{1}{2}\, b \cdot \int |V|^2 \nabla\phi.$$

Substituting gives the result.
□

4. Renormalized Caccioppoli estimate

4.1. **Lemma 5 (Renormalized Caccioppoli estimate).** Let V, P solve the renormalized equation on $B_R \times (\tau_1, \tau_2)$, where $0 < r \le R/2$. Let

$$I_R = (\tau_1, \tau_2), \qquad Q_R = B_R \times I_R.$$

Choose σ_1, σ_2 with

$$\tau_1 < \sigma_1 < \sigma_2 < \tau_2,$$

and set

$$I_r = [\sigma_1, \sigma_2], \qquad Q_r = B_r \times I_r.$$

Assume:

- V, P are classical on Q_R,
- $a, b \in L^\infty(I_R)$.

Then there exists $C > 0$, depending only on dimension and cutoff choices, such that

$$\begin{aligned}\nu \iint_{Q_r} |\nabla V|^2 \le{}& \frac{C}{\sigma_2 - \sigma_1} \iint_{Q_R} |V|^2 + \frac{C\nu}{(R-r)^2} \iint_{Q_R} |V|^2 \\ &+ \frac{C}{R-r} \iint_{Q_R} |V|^3 + \frac{C}{R-r} \iint_{Q_R} |P||V| \\ &+ C\|a\|_{L^\infty(I_R)} \iint_{Q_R} |V|^2 + \frac{C\|b\|_{L^\infty(I_R)}}{R-r} \iint_{Q_R} |V|^2.\end{aligned}$$

If in addition $a(\tau) \ge 0$ on I_R, then

$$\begin{aligned}\nu \iint_{Q_r} |\nabla V|^2 + \iint_{Q_r} a(\tau)\frac{|V|^2}{2} \le{}& \frac{C}{\sigma_2 - \sigma_1} \iint_{Q_R} |V|^2 + \frac{C\nu}{(R-r)^2} \iint_{Q_R} |V|^2 \\ &+ \frac{C}{R-r} \iint_{Q_R} |V|^3 + \frac{C}{R-r} \iint_{Q_R} |P||V| \\ &+ C\|a\|_{L^\infty(I_R)} \iint_{Q_R} |V|^2 + \frac{C\|b\|_{L^\infty(I_R)}}{R-r} \iint_{Q_R} |V|^2.\end{aligned}$$

4.1.1. *Proof.* Choose $\phi \in C_c^\infty(B_R)$ such that

$$0 \le \phi \le 1, \qquad \phi \equiv 1 \text{ on } B_r,$$

with

$$|\nabla \phi| \le \frac{C}{R-r}, \qquad |\Delta \phi| \le \frac{C}{(R-r)^2}.$$

Choose $\chi \in C_c^\infty((\tau_1, \tau_2))$ such that

$$0 \le \chi \le 1, \qquad \chi \equiv 1 \text{ on } [\sigma_1, \sigma_2],$$

and

$$|\chi'| \le \frac{C}{\sigma_2 - \sigma_1}.$$

Set

$$\zeta(y, \tau) := \phi(y)\chi(\tau).$$

Apply Lemma 3:

$$\iint_{Q_R} \nu |\nabla V|^2 \zeta + \iint_{Q_R} ae\,\zeta = I_1 + I_2 + I_3 + I_4 + I_5,$$

where

$$I_1 = \iint e\,\partial_\tau \zeta, \qquad I_2 = \nu \iint e\,\Delta\zeta,$$

$$I_3 = \iint (e+P)V\cdot\nabla\zeta, \qquad I_4 = -\iint ae\,y\cdot\nabla\zeta, \qquad I_5 = -\iint e\,b\cdot\nabla\zeta.$$

Estimate each term.

Because $|\chi'| \le C/(\sigma_2-\sigma_1)$,

$$|I_1| \le \frac{C}{\sigma_2-\sigma_1}\iint_{Q_R} |V|^2.$$

Because $|\Delta\phi| \le C/(R-r)^2$,

$$|I_2| \le \frac{C\nu}{(R-r)^2}\iint_{Q_R} |V|^2.$$

For the flux term,

$$|I_3| \le \frac{C}{R-r}\iint_{Q_R} |V|^3 + \frac{C}{R-r}\iint_{Q_R} |P||V|.$$

For the scale-modulation flux,

$$|I_4| \le \|a\|_{L^\infty(I_R)}\iint_{Q_R} e\,|y|\,|\nabla\phi|\,\chi.$$

Since $r \le R/2$, one has $\frac{R}{R-r} \le 2$, so

$$|I_4| \le C\|a\|_{L^\infty(I_R)}\iint_{Q_R} |V|^2.$$

For the translation flux,

$$|I_5| \le \frac{C\|b\|_{L^\infty(I_R)}}{R-r}\iint_{Q_R} |V|^2.$$

Since $\zeta \equiv 1$ on Q_r,

$$\nu\iint_{Q_r} |\nabla V|^2 \le \iint_{Q_R} \nu|\nabla V|^2\zeta.$$

If $a \ge 0$, then

$$\iint_{Q_r} ae \le \iint_{Q_R} ae\,\zeta.$$

Combining the previous bounds proves the desired inequalities.
□

5. Rigidity of compact low-dissipation subsequences

5.1. **Lemma 6 (Nonnegative integrable cost has a vanishing subsequence).** Let $f : [\tau_0,\infty) \to [0,\infty)$ be measurable. If

$$\int_{\tau_0}^{\infty} f(\tau)\,d\tau < \infty,$$

then there exists a sequence $\tau_n \to \infty$ such that

$$f(\tau_n) \to 0.$$

5.1.1. *Proof.* Suppose not. Then there exists $\varepsilon > 0$ and $T > \tau_0$ such that

$$f(\tau) \geq \varepsilon \qquad \text{for all } \tau \geq T.$$

Then

$$\int_{\tau_0}^{\infty} f(\tau)\, d\tau \geq \int_{T}^{\infty} \varepsilon\, d\tau = \infty,$$

contradiction.

□

5.2. **Lemma 7 (Low renormalization cost implies low core dissipation).** Fix $R_0 > 0$. Let $\phi_{R_0} \geq 0$ satisfy $\phi_{R_0} \equiv 1$ on B_{R_0}. Define

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int |V(y,\tau)|^2 \phi_{R_0}(y)\, dy,$$

where $a_+(\tau) = \max(a(\tau), 0)$.

Then for any sequence τ_n,

$$\tilde{\mathfrak{D}}_{R_0}(\tau_n) \to 0 \quad \Longrightarrow \quad \int_{B_{R_0}} |\nabla V(\tau_n, y)|^2\, dy \to 0.$$

5.2.1. *Proof.* Since $\phi_{R_0} \equiv 1$ on B_{R_0} and the second term in $\tilde{\mathfrak{D}}_{R_0}$ is nonnegative,

$$\int_{B_{R_0}} |\nabla V|^2 \leq \int |\nabla V|^2 \phi_{R_0} \leq \frac{1}{\nu} \tilde{\mathfrak{D}}_{R_0}.$$

Taking $n \to \infty$ proves the claim.

□

5.3. **Lemma 8 (Compact normalized low-dissipation subsequence is impossible).** Let $(V_n)_{n\geq 1}$ be a sequence of divergence-free vector fields on $\mathbb{R}^3$ such that:

(1) $V_n \in H^1_{\mathrm{loc}}(\mathbb{R}^3) \cap L^3(\mathbb{R}^3)$ for all n;
(2) there exists V_* such that

$$V_n \to V_* \quad \text{strongly in } L^2_{\mathrm{loc}}(\mathbb{R}^3) \cap L^3_{\mathrm{loc}}(\mathbb{R}^3);$$

(1) for every $\varepsilon > 0$, there exists $R_\varepsilon > 0$ such that

$$\sup_n \int_{|y|>R_\varepsilon} |V_n(y)|^3\, dy < \varepsilon;$$

(1) normalization:

$$\|V_n\|_{L^3(\mathbb{R}^3)} = 1 \quad \text{for all } n;$$

(1) there exists $R_0 > 0$ such that

$$\int_{B_{R_0}} |\nabla V_n(y)|^2\, dy \to 0.$$

Then such a sequence cannot exist.

5.3.1. *Proof.* By strong $L^2(B_{R_0})$-convergence, it suffices to show that $\nabla V_* = 0$ in B_{R_0}.

Take any $\Psi \in C_c^\infty(B_{R_0}; \mathbb{R}^{3\times 3})$. Then

$$\left| \int_{B_{R_0}} V_n \cdot \operatorname{div} \Psi \, dy \right| = \left| \int_{B_{R_0}} \nabla V_n : \Psi \, dy \right| \leq \|\nabla V_n\|_{L^2(B_{R_0})} \|\Psi\|_{L^2(B_{R_0})}.$$

By assumption 5, the right-hand side tends to 0. Passing to the limit using strong $L^2(B_{R_0})$-convergence yields

$$\int_{B_{R_0}} V_* \cdot \operatorname{div} \Psi \, dy = 0.$$

Hence $\nabla V_* = 0$ in $\mathcal{D}'(B_{R_0})$, so V_* is a.e. constant on B_{R_0}:

$$V_*(y) \equiv c \quad \text{on } B_{R_0}$$

for some $c \in \mathbb{R}^3$.

We claim $c = 0$. If $c \neq 0$, then for every $R < R_0$,

$$\int_{B_R} |V_n|^3 \to \int_{B_R} |c|^3 = |c|^3 |B_R|.$$

Choose $R < R_0$ large enough that $|c|^3 |B_R| > 2$. Then for all large n,

$$\int_{B_R} |V_n|^3 > 1,$$

contradicting $\|V_n\|_{L^3}^3 = 1$. Thus $c = 0$, so $V_* \equiv 0$ on B_{R_0}.

Now fix $\varepsilon > 0$. By tightness choose $R > R_0$ such that

$$\sup_n \int_{|y|>R} |V_n(y)|^3 \, dy < \varepsilon.$$

Since $V_n \to 0$ strongly in $L^3(B_R)$,

$$\int_{B_R} |V_n|^3 \, dy \to 0.$$

Therefore for all large n,

$$\|V_n\|_{L^3}^3 = \int_{B_R} |V_n|^3 \, dy + \int_{|y|>R} |V_n|^3 \, dy < 2\varepsilon.$$

Since ε is arbitrary, $\|V_n\|_{L^3} \to 0$, contradicting normalization.

□

6. Pressure and time regularity

6.1. **Lemma 9 (Local pressure estimate modulo a constant).** Let $R > 0$, fix a time τ, and suppose $V(\cdot, \tau) \in L^3(\mathbb{R}^3; \mathbb{R}^3)$ is divergence free. Let $P(\cdot, \tau)$ solve

$$-\Delta P = \partial_i \partial_j (V_i V_j) \quad \text{in } \mathbb{R}^3.$$

Then there exists a constant $c_R(\tau) \in \mathbb{R}$ such that

$$\|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C \Big(\|V(\tau)\|_{L^3(B_{4R})}^2 + R^2 \sup_{x \in B_R} \int_{|z|>2R} \frac{|V(z,\tau)|^2}{|x-z|^5} \, dz \Big).$$

6.1.1. *Proof.* Choose $\eta \in C_c^\infty(B_{4R})$ with

$$0 \le \eta \le 1, \qquad \eta \equiv 1 \text{ on } B_{2R}.$$

Decompose

$$V_i V_j = \eta V_i V_j + (1-\eta) V_i V_j.$$

Define

$$P_{\mathrm{loc}} := \mathcal{R}_i \mathcal{R}_j (\eta V_i V_j), \qquad P_{\mathrm{far}} := P - P_{\mathrm{loc}}.$$

For the local part, Calderón–Zygmund gives

$$\|P_{\mathrm{loc}}\|_{L^{3/2}(\mathbb{R}^3)} \le C \|\eta V_i V_j\|_{L^{3/2}(\mathbb{R}^3)} \le C \|V\|_{L^3(B_{4R})}^2,$$

hence

$$\|P_{\mathrm{loc}}\|_{L^{3/2}(B_R)} \le C \|V\|_{L^3(B_{4R})}^2.$$

For the far part, observe that

$$-\Delta P_{\mathrm{far}} = \partial_i \partial_j ((1-\eta) V_i V_j),$$

and the right-hand side vanishes in B_{2R}, so P_{far} is harmonic in B_{2R}.

Let

$$K_{ij}(x) = \partial_i \partial_j \Big(\frac{1}{4\pi |x|} \Big).$$

Up to an additive constant,

$$P_{\mathrm{far}}(x) = \int_{\mathbb{R}^3} K_{ij}(x-z)(1-\eta(z)) V_i(z) V_j(z)\, dz.$$

Fix $x_0 \in B_R$ and set $c_R := P_{\mathrm{far}}(x_0)$. Then for $x \in B_R$,

$$P_{\mathrm{far}}(x) - c_R = \int_{|z|>2R} \big(K_{ij}(x-z) - K_{ij}(x_0 - z) \big) V_i(z) V_j(z)\, dz.$$

Using the mean value theorem and $|\nabla K_{ij}(\xi)| \le C|\xi|^{-4}$,

$$|K_{ij}(x-z) - K_{ij}(x_0-z)| \le C|x-x_0||x-z|^{-4}.$$

Since $x, x_0 \in B_R$, $|x - x_0| \le 2R$, so

$$|P_{\mathrm{far}}(x) - c_R| \le CR \int_{|z|>2R} \frac{|V(z)|^2}{|x-z|^4}\, dz.$$

Because $|x-z| \ge cR$ on $|z| > 2R$, $x \in B_R$,

$$\int_{|z|>2R} \frac{|V(z)|^2}{|x-z|^4}\, dz \le CR \sup_{x \in B_R} \int_{|z|>2R} \frac{|V(z)|^2}{|x-z|^5}\, dz.$$

Hence

$$|P_{\mathrm{far}}(x) - c_R| \le CR^2 \sup_{x \in B_R} \int_{|z|>2R} \frac{|V(z)|^2}{|x-z|^5}\, dz.$$

Taking the $L^{3/2}(B_R)$-norm gives

$$\|P_{\mathrm{far}} - c_R\|_{L^{3/2}(B_R)} \le CR^2 \sup_{x \in B_R} \int_{|z|>2R} \frac{|V(z)|^2}{|x-z|^5}\, dz.$$

Since

$$P - c_R = P_{\mathrm{loc}} + (P_{\mathrm{far}} - c_R),$$

the stated estimate follows.

□

6.2. **Lemma 9.1 (Uniform pressure-tail control from global L^3).** For $R > 0$, define

$$\mathcal{T}_R(\tau) := \sup_{x\in B_R} \int_{|z|>2R} \frac{|V(z,\tau)|^2}{|x-z|^5}\, dz.$$

If

$$M_3 := \sup_{\tau\geq\tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then

$$\sup_{\tau\geq\tau_0} \mathcal{T}_R(\tau) \leq CR^{-4}M_3^2.$$

6.2.1. *Proof.* Fix $x \in B_R$ and decompose

$$\{|z| > 2R\} = \bigcup_{k=1}^{\infty} A_k, \qquad A_k := \{2^k R < |z| \leq 2^{k+1}R\}.$$

For $z \in A_k$,

$$|x - z| \geq |z| - |x| \geq 2^k R - R \geq 2^{k-1}R.$$

Therefore

$$\int_{|z|>2R} \frac{|V(z,\tau)|^2}{|x-z|^5}\, dz \leq C \sum_{k\geq 1} (2^k R)^{-5} \int_{A_k} |V(z,\tau)|^2\, dz.$$

By Holder,

$$\int_{A_k} |V|^2 \leq |A_k|^{1/3} \left(\int_{A_k} |V|^3\right)^{2/3} \leq C\, 2^k R\, a_k(\tau)^{2/3},$$

where

$$a_k(\tau) := \int_{A_k} |V(z,\tau)|^3\, dz.$$

Thus

$$\mathcal{T}_R(\tau) \leq CR^{-4} \sum_{k\geq 1} 2^{-4k} a_k(\tau)^{2/3}.$$

Using Holder on the counting measure with exponents 3 and 3/2,

$$\sum_{k\geq 1} 2^{-4k} a_k^{2/3} \leq \left(\sum_{k\geq 1} 2^{-12k}\right)^{1/3} \left(\sum_{k\geq 1} a_k\right)^{2/3} \leq C\|V(\tau)\|^2_{L^3(\mathbb{R}^3)}.$$

Taking the supremum over $x \in B_R$ and $\tau \geq \tau_0$ proves the claim.
□

6.3. **Lemma 9.2 (Verification of the local L^2 pressure hypotheses).** Let $I \subset \mathbb{R}$ be bounded and $R > 0$. Assume

$$\sup_{\tau\in I} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq M_3$$

and

$$V \in L^2(I; H^1(B_{4R})).$$

Then

$$V \in L^4(I; L^4(B_{4R})).$$

Moreover, if

$$\mathcal{H}_R(\tau) := \sup_{x\in B_R} \int_{|z|>2R} \frac{|V(z,\tau)|^2}{|x-z|^4}\, dz,$$

then

$$\|\mathcal{H}_R\|_{L^\infty(I)} \le CR^{-3}M_3^2, \qquad \|\mathcal{H}_R\|_{L^2(I)} \le C|I|^{1/2}R^{-3}M_3^2.$$

6.3.1. *Proof.* For almost every $\tau \in I$, interpolation and Sobolev give

$$\|V(\tau)\|_{L^4(B_{4R})} \le \|V(\tau)\|_{L^3(B_{4R})}^{1/2}\|V(\tau)\|_{L^6(B_{4R})}^{1/2} \le C_R M_3^{1/2}\|V(\tau)\|_{H^1(B_{4R})}^{1/2}.$$

Thus

$$\|V(\tau)\|_{L^4(B_{4R})}^4 \le C_R M_3^2\|V(\tau)\|_{H^1(B_{4R})}^2.$$

Integrating in τ proves $V \in L^4(I; L^4(B_{4R}))$.

For $\mathcal{H}_R$, use the same dyadic annuli A_k as in Lemma 9.1. For $x \in B_R$,

$$\int_{|z|>2R} \frac{|V(z,\tau)|^2}{|x-z|^4}\,dz \le C\sum_{k\ge 1}(2^kR)^{-4}\int_{A_k}|V(z,\tau)|^2\,dz.$$

The previous Holder estimate gives

$$\int_{A_k}|V|^2 \le C\,2^kR\,a_k(\tau)^{2/3},$$

so

$$\mathcal{H}_R(\tau) \le CR^{-3}\sum_{k\ge 1}2^{-3k}a_k(\tau)^{2/3}.$$

Again,

$$\sum_{k\ge 1}2^{-3k}a_k(\tau)^{2/3} \le \left(\sum_{k\ge 1}2^{-9k}\right)^{1/3}\left(\sum_{k\ge 1}a_k(\tau)\right)^{2/3} \le C\|V(\tau)\|_{L^3(\mathbb{R}^3)}^2.$$

Hence $\mathcal{H}_R(\tau) \le CR^{-3}M_3^2$, which gives both asserted bounds.

□

6.4. **Lemma 9.3 (Local L^2 pressure estimate modulo constants).** Let $I \subset \mathbb{R}$ be bounded. Assume

$$V \in L^4(I; L^4(B_{4R})), \qquad \mathcal{H}_R \in L^2(I),$$

where $\mathcal{H}_R$ is defined in Lemma 9.2. Suppose

$$-\Delta P = \partial_i\partial_j(V_iV_j) \quad \text{in } \mathbb{R}^3$$

for almost every $\tau \in I$. Then there exists a measurable $c_R : I \to \mathbb{R}$ such that

$$P - c_R \in L^2(I; L^2(B_R)),$$

and

$$\|P - c_R\|_{L^2(I;L^2(B_R))} \le C_R\left(\|V\|_{L^4(I;L^4(B_{4R}))}^2 + \|\mathcal{H}_R\|_{L^2(I)}\right).$$

6.4.1. *Proof.* Choose $\eta \in C_c^\infty(B_{4R})$ with $0 \le \eta \le 1$ and $\eta \equiv 1$ on B_{2R}. Write

$$V_iV_j = \eta V_iV_j + (1-\eta)V_iV_j$$

and define

$$P_{\mathrm{loc}} := \mathcal{R}_i\mathcal{R}_j(\eta V_iV_j), \qquad P_{\mathrm{far}} := P - P_{\mathrm{loc}}.$$

Calderon-Zygmund on $L^2(\mathbb{R}^3)$ gives

$$\|P_{\mathrm{loc}}(\tau)\|_{L^2(\mathbb{R}^3)} \le C\|V(\tau)\|^2_{L^4(B_{4R})},$$

hence

$$\|P_{\mathrm{loc}}\|_{L^2(I;L^2(B_R))} \le C\|V\|^2_{L^4(I;L^4(B_{4R}))}.$$

For the far term, P_{far} is harmonic in B_{2R} modulo constants. Let

$$K_{ij}(x) := \partial_i\partial_j\left(\frac{1}{4\pi|x|}\right).$$

Fix $x_0 = 0$. Since pressure is defined modulo constants, choose $c_R(\tau)$ so that for $x \in B_R$,

$$P_{\mathrm{far}}(x,\tau) - c_R(\tau) = \int_{|z|>2R}\big(K_{ij}(x-z) - K_{ij}(x_0-z)\big)(1-\eta(z))V_i(z,\tau)V_j(z,\tau)\,dz.$$

The kernel estimate

$$|\nabla K_{ij}(\xi)| \le C|\xi|^{-4}$$

and the mean value theorem imply

$$|K_{ij}(x-z) - K_{ij}(x_0-z)| \le CR|x-z|^{-4}$$

for $x \in B_R$, $|z| > 2R$. Hence

$$|P_{\mathrm{far}}(x,\tau) - c_R(\tau)| \le CR\,\mathcal{H}_R(\tau).$$

Taking the $L^2(B_R)$-norm gives

$$\|P_{\mathrm{far}}(\tau) - c_R(\tau)\|_{L^2(B_R)} \le CR^{5/2}\mathcal{H}_R(\tau).$$

Taking $L^2(I)$ and adding the local estimate proves the result.
□

6.5. **Lemma 10 (Local $L^2_\tau H^{-1}_x$ time-regularity).** Let $R > 0$, and let V, P solve the renormalized equation on $B_{2R} \times I$, $I \subset \mathbb{R}$. Assume:

(1) local H^1-bound:

$$\sup_{\tau\in I}\|V(\tau)\|_{H^1(B_{2R})} \le M_1;$$

(1) local L^2 pressure bound modulo constants:

$$\text{there exists } c \in L^2(I) \text{ such that } \|P - c\|_{L^2(I;L^2(B_{2R}))} \le M_{P,2};$$

(1) bounded modulation parameters:

$$\sup_{\tau\in I}\big(|a(\tau)| + |b(\tau)|\big) \le M_{ab}.$$

Then there exists $C = C(R,\nu)$ such that

$$\|\partial_\tau V\|_{L^2(I;H^{-1}(B_R))} \le C\Big(|I|^{1/2}\big(M_1 + M_1^2 + M_{ab}(M_1+1)\big) + M_{P,2}\Big).$$

In particular, by Lemmas 9.2 and 9.3, hypothesis 2 follows on bounded intervals from

$$\sup_{\tau\in I}\|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty, \qquad V \in L^2(I;H^1(B_{8R})),$$

and
$$-\Delta P = \partial_i \partial_j (V_i V_j) \quad \text{in } \mathbb{R}^3$$
for almost every $\tau \in I$.

6.5.1. *Proof.* We estimate each term in
$$\partial_\tau V = \nu \Delta V - (V \cdot \nabla) V - \nabla P + a(\tau) V + a(\tau)\, y \cdot \nabla V + b(\tau) \cdot \nabla V$$
in $H^{-1}(B_R)$.

Let $\varphi \in H_0^1(B_R)$ with $\|\varphi\|_{H^1(B_R)} \leq 1$.

Laplacian term.
$$\langle \Delta V, \varphi \rangle = -\int_{B_R} \nabla V : \nabla \varphi,$$
hence
$$\|\Delta V\|_{H^{-1}(B_R)} \leq \|\nabla V\|_{L^2(B_R)} \leq M_1.$$

Pressure term. Choose $c \in L^2(I)$ as in hypothesis 2. Replacing P by $P - c(\tau)$ does not change ∇P. Thus, for almost every $\tau \in I$,
$$\langle \nabla P, \varphi \rangle = -\int_{B_R} (P - c(\tau))\, \nabla \cdot \varphi.$$
Since
$$\|\nabla \cdot \varphi\|_{L^2(B_R)} \leq \|\nabla \varphi\|_{L^2(B_R)} \leq \|\varphi\|_{H^1(B_R)},$$
we have
$$\|\nabla P\|_{H^{-1}(B_R)} \leq \|P(\tau) - c(\tau)\|_{L^2(B_R)}.$$
Taking $L^2(I)$ gives
$$\|\nabla P\|_{L^2(I; H^{-1}(B_R))} \leq M_{P,2}.$$

Transport term. Since $\nabla \cdot V = 0$,
$$(V \cdot \nabla) V = \nabla \cdot (V \otimes V),$$
hence
$$\langle (V \cdot \nabla) V, \varphi \rangle = -\int_{B_R} (V \otimes V) : \nabla \varphi.$$
Therefore
$$\|(V \cdot \nabla) V\|_{H^{-1}(B_R)} \leq \|V \otimes V\|_{L^2(B_R)} = \|V\|_{L^4(B_R)}^2.$$
By bounded-domain Gagliardo–Nirenberg and Sobolev,
$$\|V\|_{L^4(B_R)}^2 \leq C(R) \|V\|_{H^1(B_R)}^2 \leq C(R) M_1^2.$$

The aV term.
$$|\langle aV, \varphi \rangle| \leq |a| \|V\|_{L^2(B_R)} \|\varphi\|_{L^2(B_R)} \leq C(R) M_{ab} M_1.$$

The $a\, y \cdot \nabla V$ term.
$$\langle y \cdot \nabla V, \varphi \rangle = \sum_{j=1}^{3} \int_{B_R} y_j \partial_j V \cdot \varphi.$$
Integrating by parts,
$$\int_{B_R} y_j \partial_j V \cdot \varphi = -\int_{B_R} V \cdot \partial_j (y_j \varphi) = -\int_{B_R} V \cdot \varphi - \int_{B_R} y_j V \cdot \partial_j \varphi.$$
Hence
$$|\langle y \cdot \nabla V, \varphi \rangle| \leq 3 \|V\|_{L^2(B_R)} \|\varphi\|_{L^2(B_R)} + R \|V\|_{L^2(B_R)} \|\nabla \varphi\|_{L^2(B_R)},$$

so

$$\|y \cdot \nabla V\|_{H^{-1}(B_R)} \le C(R)M_1.$$

Therefore

$$\|a(\tau)\, y \cdot \nabla V\|_{H^{-1}(B_R)} \le C(R)M_{ab}M_1.$$

The $b \cdot \nabla V$ term.

$$\langle b \cdot \nabla V, \varphi\rangle = -\int_{B_R} V \cdot (b \cdot \nabla\varphi),$$

hence

$$\|b \cdot \nabla V\|_{H^{-1}(B_R)} \le |b|\|V\|_{L^2(B_R)} \le M_{ab}M_1.$$

Combining the six estimates yields

$$\|\partial_\tau V\|_{L^2(I;H^{-1}(B_R))} \le C\Big(|I|^{1/2}\big(M_1 + M_1^2 + M_{ab}(M_1+1)\big) + M_{P,2}\Big).$$

□

7. Aubin–Lions–Simon closure

7.1. **Lemma 11 (Aubin–Lions–Simon closure of the low-cost sequence).** Assume:

(1) For every $R > 0$,

$$\sup_{\tau\ge\tau_0} \|V(\tau)\|_{H^1(B_{2R})} \le M_1(R).$$

(1) For every $R > 0$,

$$\sup_{T\ge\tau_0} \|\partial_\tau V\|_{L^2((T,T+1);H^{-1}(B_R))} \le M_2(R).$$

(1) The orbit is uniformly L^3-tight and L^3-normalized:

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} = 1,$$

and

$$\forall\varepsilon > 0\ \exists R_\varepsilon :\ \sup_{\tau\ge\tau_0} \int_{|y|>R_\varepsilon} |V(y,\tau)|^3 < \varepsilon.$$

(1) There exists $\tau_n \to \infty$ such that

$$\tilde{\mathfrak{D}}_{R_0}(\tau_n) \to 0.$$

Then, after passing to a subsequence, there exists V_* such that

$$V(\tau_n) \to V_* \quad \text{strongly in } L^2_{\text{loc}}(\mathbb{R}^3) \cap L^3_{\text{loc}}(\mathbb{R}^3),$$

and

$$\int_{B_{R_0}} |\nabla V(\tau_n)|^2 \to 0.$$

Consequently, Theorem 8 applies and gives a contradiction.

7.1.1. *Proof.* By Lemma 7,

$$\tilde{\mathfrak{D}}_{R_0}(\tau_n) \to 0 \quad \Longrightarrow \quad \int_{B_{R_0}} |\nabla V(\tau_n)|^2 \to 0.$$

It remains to obtain strong local compactness of $V(\tau_n)$. Fix $R > 0$, and define shifted trajectories

$$W_n(s,y) := V(\tau_n + s, y), \qquad s \in (0,1),\ y \in B_R.$$

By assumption 1,

$$\sup_n \|W_n\|_{L^\infty(0,1;H^1(B_R))} < \infty,$$

hence

$$\sup_n \|W_n\|_{L^2(0,1;H^1(B_R))} < \infty.$$

By assumption 2,

$$\sup_n \|\partial_s W_n\|_{L^2(0,1;H^{-1}(B_R))} < \infty.$$

Since

$$H^1(B_R) \hookrightarrow\hookrightarrow L^2(B_R) \hookrightarrow H^{-1}(B_R),$$

the Aubin–Lions–Simon theorem implies, after passage to a subsequence,

$$W_n \to W_* \quad \text{strongly in } L^2(0,1;L^2(B_R)).$$

By Lions–Magenes trace regularity,

$$W_n \in C([0,1];L^2(B_R)),$$

with uniform control. By Simon's compactness theorem in the time-continuous version, after a further subsequence,

$$W_n \to W_* \quad \text{strongly in } C([0,1];L^2(B_R)).$$

In particular, at time $s = 0$,

$$V(\tau_n,\cdot) = W_n(0,\cdot) \to W_*(0,\cdot) \quad \text{strongly in } L^2(B_R).$$

Since $R > 0$ was arbitrary, a diagonal extraction gives a subsequence and a function V_* such that

$$V(\tau_n) \to V_* \quad \text{strongly in } L^2_{\text{loc}}(\mathbb{R}^3).$$

Now, from the local H^1-bound and strong L^2-convergence on each B_R, interpolation yields strong L^3-convergence on each B_R:

$$\|V(\tau_n) - V_*\|_{L^3(B_R)} \le \|V(\tau_n) - V_*\|_{L^2(B_R)}^{1/2} \|V(\tau_n) - V_*\|_{L^6(B_R)}^{1/2}.$$

By Sobolev, $H^1(B_R) \hookrightarrow L^6(B_R)$, and the H^1-bound is uniform, so the right-hand side tends to 0. Hence

$$V(\tau_n) \to V_* \quad \text{strongly in } L^3_{\text{loc}}(\mathbb{R}^3).$$

With uniform L^3-tightness and normalization, all the assumptions of Lemma 8 are satisfied. Since the low core dissipation was already shown above, Lemma 8 gives a contradiction.

□

8. Final exclusion theorem

8.1. **Theorem A (compact Type II renormalization criterion).** Let V, P solve the renormalized Navier–Stokes equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)\big(V + y \cdot \nabla V\big) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0$$

on $[\tau_0, \infty) \times \mathbb{R}^3$.

Fix $R_0 > 0$ and a cutoff $\phi_{R_0} \in C_c^\infty(\mathbb{R}^3)$, $\phi_{R_0} \geq 0$, $\phi_{R_0} \equiv 1$ on B_{R_0}, and define

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\, dy.$$

Assume:

8.1.1. *(H1) global L^3-normalization.*

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} = 1 \qquad \forall \tau \geq \tau_0.$$

8.1.2. *(H2) uniform L^3-tightness.* For every $\varepsilon > 0$, there exists $R_\varepsilon > 0$ such that

$$\sup_{\tau \geq \tau_0} \int_{|y| > R_\varepsilon} |V(y,\tau)|^3\, dy < \varepsilon.$$

8.1.3. *(H3) uniform local H^1-bounds.* For every $R > 0$, there exists $M_1(R)$ such that

$$\sup_{\tau \geq \tau_0} \|V(\tau)\|_{H^1(B_{2R})} \leq M_1(R).$$

8.1.4. *(H4) uniform boundedness of the modulation parameters.* There exists M_{ab} such that

$$\sup_{\tau \geq \tau_0} \big(|a(\tau)| + |b(\tau)|\big) \leq M_{ab}.$$

8.1.5. *Derived pressure and time-regularity input.* By (H1), Lemma 9.1 gives, for every $R > 0$,

$$\sup_{\tau \geq \tau_0} \sup_{x \in B_R} \int_{|z| > 2R} \frac{|V(z,\tau)|^2}{|x - z|^5}\, dz \leq C R^{-4}.$$

By (H1), (H3), (H4), and Lemmas 9.2, 9.3, and 10, for every $R > 0$,

$$\sup_{T \geq \tau_0} \|\partial_\tau V\|_{L^2((T,T+1);H^{-1}(B_R))} < \infty.$$

Then

$$\int_{\tau_0}^\infty \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

8.1.6. *Proof.* Assume for contradiction that

$$\int_{\tau_0}^\infty \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau < \infty.$$

By Lemma 6, there exists a sequence $\tau_n \to \infty$ such that

$$\tilde{\mathfrak{D}}_{R_0}(\tau_n) \to 0.$$

By Lemma 11, after passing to a subsequence,

$$V(\tau_n) \to V_* \quad \text{strongly in } L^2_{\mathrm{loc}}(\mathbb{R}^3) \cap L^3_{\mathrm{loc}}(\mathbb{R}^3),$$

and

$$\int_{B_{R_0}} |\nabla V(\tau_n)|^2 \to 0.$$

Then Lemma 8 applies and yields a contradiction.

Therefore

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

□

9. Repaired scale gauge

9.1. **G1. Repaired weighted-moment scale gauge and centering.**

Keep the centering gauge

$$G_j(V) := \int_{\mathbb{R}^3} y_j |V(y)|^2 \psi_R(y)\, dy = 0, \qquad j = 1, 2, 3,$$

with $\psi_R \in C_c^\infty(\mathbb{R}^3)$, $\psi_R \equiv 1$ on B_R, and $\psi_R \equiv 0$ outside B_{2R}.

Set the scale gauge as

$$G_0(V) := \int_{\mathbb{R}^3} |y|^{-p} |V(y)|^3\, dy - \Theta_0 = 0, \qquad 0 < p < 3,\ \Theta_0 > 0.$$

In physical variables this is

$$\lambda(t)^p \int_{\mathbb{R}^3} |x - x_c(t)|^{-p} |u(x,t)|^3\, dx = \Theta_0.$$

The scale-fixing condition used below is the weighted-moment gauge above.

9.2. **G2. Derivatives and exact scale derivative.**

Let W be a test variation and $Z_{\mathrm{sc}}(V) := V + y \cdot \nabla V$. Under the admissibility hypotheses in

one has

$$DG_0(V)[W] = 3 \int_{\mathbb{R}^3} |y|^{-p} |V|\, V \cdot W\, dy,$$

$$DG_0(V)[Z_{\mathrm{sc}}(V)] = p \int_{\mathbb{R}^3} |y|^{-p} |V(y)|^3\, dy.$$

Hence on the gauge surface $G_0(V) = 0$,

$$DG_0(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0 > 0.$$

For the centering rows,

$$DG_j(V)[W] = 2 \int_{\mathbb{R}^3} y_j V \cdot W\, \psi_R\, dy, \qquad j = 1, 2, 3.$$

$$DG_j(V)[Z_{\mathrm{sc}}(V)] = 2 \int_{\mathbb{R}^3} y_j V \cdot (V + y \cdot \nabla V)\, \psi_R\, dy.$$

9.3. **G3. Modulation matrix entries.**

Let

$$Z_{\mathrm{sc}}(V) := V + y \cdot \nabla V.$$

The modulation matrix $M(V) \in \mathbb{R}^{4\times 4}$ is defined by

$$M(V)_{00} = DG_0(V)[Z_{\mathrm{sc}}(V)], \quad M(V)_{0\ell} = DG_0(V)[\partial_\ell V],$$
$$M(V)_{j0} = DG_j(V)[Z_{\mathrm{sc}}(V)], \quad M(V)_{j\ell} = DG_j(V)[\partial_\ell V].$$

Hence

$$M_{00}(V) = p \int_{\mathbb{R}^3} |y|^{-p} |V(y)|^3\, dy = p\Theta_0,$$

$$M_{0\ell}(V) = 3 \int_{\mathbb{R}^3} |y|^{-p} |V|\, V \cdot \partial_\ell V\, dy = \int_{\mathbb{R}^3} |y|^{-p} \partial_\ell(|V|^3)\, dy = p \int_{\mathbb{R}^3} y_\ell |y|^{-p-2} |V(y)|^3\, dy,$$

$$M_{j0}(V) = 2\int_{\mathbb{R}^3} y_j\, V\cdot(V + y\cdot\nabla V)\,\psi_R\,dy,$$
$$M_{j\ell}(V) = -\delta_{j\ell}\int_{\mathbb{R}^3} |V|^2\psi_R - \int_{\mathbb{R}^3} |V|^2 y_j\partial_\ell\psi_R\,dy.$$

The translation block is unchanged from the standard cutoff form; the scale row is replaced by the weighted formulas above.

9.4. **G4. Continuity of the modulation matrix.** If

$$V_n \to V \quad \text{in } W^{1,3}(B_{2R}) \cap H^1(B_{2R}),$$

then

$$M(V_n) \to M(V) \quad \text{in } \mathbb{R}^{4\times 4}.$$

9.4.1. *Proof.* Each entry of $M(V)$ is an integral of one of the following forms:

- $\int |V|^3\omega$,
- $\int |V|^2\omega$,
- $\int y_j V\cdot y\cdot\nabla V\,\omega$, with $\omega \in C_c^\infty(B_{2R})$.

If $V_n \to V$ in $L^3(B_{2R})$, then $|V_n|^3 \to |V|^3$ in L^1. If $V_n \to V$ in $L^2(B_{2R})$, then $|V_n|^2 \to |V|^2$ in L^1. If moreover $\nabla V_n \to \nabla V$ in $L^3(B_{2R})$, then the mixed terms with $y\cdot\nabla V_n$ also converge in L^1.

Therefore each matrix entry converges, hence the whole matrix converges.

□

9.5. **G5. Translation block invertibility under core-mass dominance.** Let

$$M_{\mathrm{tr}}(V) := (M_{j\ell}(V))_{1\le j,\ell\le 3}.$$

Assume:

- ψ_R is radial, nonincreasing, $\psi_R \equiv 1$ on B_R, supported in B_{2R};
- there exist $m_0 > 0$, $\varepsilon_0 > 0$ such that

$$\int |V|^2\psi_R \ge m_0, \qquad \int_{A_R} |V|^2 \le \varepsilon_0,$$

where $A_R = \{R \le |y| \le 2R\}$;

- ε_0 is small enough relative to m_0.

Then $M_{\mathrm{tr}}(V)$ is invertible and

$$\|M_{\mathrm{tr}}(V)^{-1}\| \le C(m_0, R).$$

9.5.1. *Proof.* From the explicit formula,

$$M_{j\ell}(V) = -\delta_{j\ell}\int |V|^2\psi_R - \int |V|^2 y_j\partial_\ell\psi_R.$$

Write

$$m_\psi(V) := \int |V|^2\psi_R.$$

Then

$$M_{\mathrm{tr}}(V) = -m_\psi(V) I_3 + E(V),$$

where

$$E_{j\ell}(V) := -\int |V|^2 y_j\partial_\ell\psi_R.$$

Since $\partial_\ell\psi_R$ is supported in A_R,

$$|E_{j\ell}(V)| \le C_R\int_{A_R} |V|^2 \le C_R\varepsilon_0.$$

Thus

$$\|E(V)\| \leq C_R\varepsilon_0.$$

Because $m_\psi(V) \geq m_0$,

$$M_{\mathrm{tr}}(V) = -m_\psi(V)\left(I_3 - m_\psi(V)^{-1}E(V)\right).$$

If $C_R\varepsilon_0 \leq m_0/2$, then

$$\|m_\psi(V)^{-1}E(V)\| \leq \frac{1}{2},$$

so $I_3 - m_\psi(V)^{-1}E(V)$ is invertible by the Neumann series, and

$$\|M_{\mathrm{tr}}(V)^{-1}\| \leq \frac{2}{m_0}.$$

□

9.6. **G6. Full matrix invertibility by Schur complement under quantitative hypotheses.** Write the full modulation matrix as

$$M(V) = \begin{pmatrix} \alpha(V) & u(V)^T \\ v(V) & A(V) \end{pmatrix},$$

where

- $\alpha(V) = M_{00}(V)$,
- $A(V) = M_{\mathrm{tr}}(V)$,
- $u, v \in \mathbb{R}^3$ are the cross terms.

Assume along the orbit:

(1) $\alpha(V) \geq \alpha_0 > 0$,
(2) $A(V)$ is invertible with

$$\|A(V)^{-1}\| \leq C_A,$$

(1) the Schur complement is uniformly positive:

$$\alpha(V) - u(V)^T A(V)^{-1} v(V) \geq \sigma_0 > 0.$$

Then $M(V)$ is invertible and

$$\|M(V)^{-1}\| \leq C(\alpha_0, \sigma_0, C_A, \|u\|, \|v\|).$$

9.6.1. *Proof.* This is the standard Schur-complement formula. Since A is invertible,

$$M^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}u^T A^{-1} \\ -A^{-1}vS^{-1} & A^{-1} + A^{-1}vS^{-1}u^T A^{-1} \end{pmatrix},$$

where

$$S = \alpha - u^T A^{-1} v.$$

Since $S \geq \sigma_0 > 0$, the formula is well-defined and the inverse is uniformly bounded in terms of the displayed quantities.

□

9.7. **G7. Bounded modulation parameters under uniform nondegeneracy.** Let X be a Banach space continuously embedded in

$$H^2(B_{2R}) \cap W^{1,3}(B_{2R}) \cap L^3(B_{2R}).$$

Assume:

- $\sup_\tau \|V(\tau)\|_X \le M_X$,
- the gauge derivatives are uniformly bounded on the orbit,
- the full matrix $M(V(\tau))$ is uniformly invertible,
- the local L^2 pressure bound of Lemma 9.3 holds on the windows used to differentiate the gauge constraints.

Then

$$\sup_\tau \big(|a(\tau)| + |b(\tau)|\big) < \infty.$$

9.7.1. *Proof.* Differentiate the gauge constraints $G_k(V(\tau)) = 0$:

$$0 = DG_k(V)\big[\nu\Delta V - (V\cdot\nabla)V - \nabla P\big] + a\, DG_k(V)[Z_{\mathrm{sc}}(V)] + \sum_{j=1}^{3} b_j DG_k(V)[\partial_j V].$$

This gives

$$M(V(\tau)) \begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = -F(V(\tau)),$$

where $F(V)$ is the forcing vector built from the non-modulation terms.

By the assumed orbit bound, the nonlinear estimate, and the pressure bound, $\|F(V(\tau))\|$ is uniformly bounded. Uniform invertibility of $M(V(\tau))$ then gives a uniform bound on $(a(\tau), b(\tau))$. $\square$

10. Analytic inputs and conclusions

10.1. **Internal analytic results.** The following statements are fully proved from the assumptions stated in each theorem:

(1) Renormalized equation (Lemma 1).
(2) Pointwise renormalized local energy identity (Lemma 2).
(3) Spacetime cutoff energy identity (Lemma 3).
(4) Localized renormalized energy identity (Lemma 4).
(5) Renormalized Caccioppoli estimate (Lemma 5).
(6) Nonnegative integrable cost has a vanishing subsequence (Lemma 6).
(7) Low cost implies low core dissipation (Lemma 7).
(8) Compact normalized low-dissipation subsequence is impossible (Lemma 8).
(9) Local $L^{3/2}$ pressure estimate modulo a constant (Lemma 9).
(10) Uniform pressure-tail control from global L^3 (Lemma 9.1).
(11) Verification of the local L^2 pressure hypotheses (Lemma 9.2).
(12) Local L^2 pressure estimate modulo constants (Lemma 9.3).
(13) Local $L^2_\tau H^{-1}_x$ time-regularity estimate (Lemma 10).
(14) Aubin–Lions–Simon closure and final contradiction under hypotheses (Lemma 11 / Theorem A).
(15) Explicit formulas and continuity for the specific gauge (G2–G4).
(16) Translation-block invertibility under quantitative core-mass dominance (G5).
(17) Full matrix invertibility by Schur complement under quantitative hypotheses (G6).
(18) Bounded modulation parameters under uniform nondegeneracy (G7).

10.2. **Auxiliary analytic inputs.** The remaining substantive analytic tasks for an actual 3D NS instantiation are:

(1) Prove the renormalized orbit satisfies the local H^1-bounds required in the chosen final theorem.
(2) Verify the cross-term smallness/Schur-complement hypotheses for the repaired scale gauge.
(3) Verify the full modulation matrix is uniformly nondegenerate along the renormalized orbit.

These are the analytic inputs used in the compactness argument.

10.3. **Summary.** The principal conclusion may be summarized as follows.

If a compact normalized renormalized Type II orbit exists and satisfies:
- uniform local H^1-bounds,
- uniform L^3-tightness,
- uniform modulation bounds,

then its renormalization cost has infinite total accumulation:

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

In particular, the orbit cannot admit a low-cost subsequence.

This is the compact Type II renormalization criterion proved in this paper.

11. Final exclusion theorem with spacetime H^1 control

11.1. **Spacetime local energy control.** The compactness argument is naturally formulated using the spacetime local energy hypothesis

$$\sup_{\tau \geq \tau_0} \|V(\tau)\|_{H^1(B_{2R})} \leq M_1(R).$$

The renormalized Caccioppoli estimate provides a **space-time** bound of the form

$$V \in L^2_{\text{loc}}\big((\tau_0, \infty); H^1_{\text{loc}}(\mathbb{R}^3)\big),$$

not an $L^\infty_\tau H^1_y$ bound.

For the Aubin–Lions–Simon compactness step, the $L^2_\tau H^1_y$ bound is the natural hypothesis and is sufficient.

Accordingly, the final theorem is stated with the $L^2_\tau H^1_y$ assumption.

11.2. **Lemma 12 (Local $L^2_\tau H^1_y$ bound from the renormalized Caccioppoli estimate).** Fix $0 < r \leq R/2$, and let

$$Q_R = B_R \times (\tau_1, \tau_2), \qquad Q_r = B_r \times [\sigma_1, \sigma_2],$$

with $\tau_1 < \sigma_1 < \sigma_2 < \tau_2$.

Assume V, P solve the renormalized equation on Q_R, and that on Q_R one has

$$\sup_{\tau \in (\tau_1, \tau_2)} \|V(\tau)\|_{L^2(B_R)} \leq M_2,$$

$$\iint_{Q_R} |V|^3 \leq M_3,$$

$$\iint_{Q_R} |P||V| \leq M_{PV},$$

$$\|a\|_{L^\infty(\tau_1,\tau_2)} + \|b\|_{L^\infty(\tau_1,\tau_2)} \le M_{ab}.$$

Then

$$\nu \iint_{Q_r} |\nabla V|^2 \le C\Big(M_2^2 + \frac{\nu}{(R-r)^2} M_2^2(\tau_2-\tau_1) + \frac{1}{R-r} M_3 + \frac{1}{R-r} M_{PV} + M_{ab} M_2^2(\tau_2-\tau_1) + \frac{M_{ab}}{R-r} M_2^2(\tau_2-\tau_1)\Big),$$

for a constant C depending only on dimension and cutoff choices.

In particular,

$$V \in L^2_{\mathrm{loc}}\big((\tau_0,\infty); H^1_{\mathrm{loc}}(\mathbb{R}^3)\big).$$

11.2.1. *Proof.* Apply Lemma 5 (the renormalized Caccioppoli estimate). The only term needing simplification is

$$\iint_{Q_R} |V|^2,$$

which is bounded by

$$\iint_{Q_R} |V|^2 \le (\tau_2-\tau_1) \sup_{\tau\in(\tau_1,\tau_2)} \|V(\tau)\|^2_{L^2(B_R)} \le (\tau_2-\tau_1) M_2^2.$$

Substituting this into Lemma 5 gives the claimed inequality.

The local $L^2_\tau H^1_y$ membership follows by applying the estimate on arbitrary compact cylinders $Q_r \Subset Q_R$.

□

11.3. **Lemma 13 (Aubin–Lions–Simon compactness with spacetime hypotheses).** Let $R > 0$, and for each n define the shifted renormalized trajectory

$$W_n(s,y) := V(\tau_n + s, y), \qquad (s,y) \in (0,1)\times B_R.$$

Assume:

(1) $V \in L^2_{\mathrm{loc}}\big((\tau_0,\infty); H^1_{\mathrm{loc}}(\mathbb{R}^3)\big)$, and for each fixed R,

$$\sup_n \|W_n\|_{L^2(0,1;H^1(B_R))} < \infty;$$

(1) for each fixed R,

$$\sup_n \|\partial_s W_n\|_{L^2(0,1;H^{-1}(B_R))} < \infty.$$

Then, after passing to a subsequence,

$$W_n \to W_* \quad \text{strongly in } L^2(0,1;L^2(B_R)).$$

If in addition one has a Lions–Magenes trace bound so that

$$W_n \in C([0,1];L^2(B_R))$$

uniformly, then after passing to a further subsequence,

$$W_n(0,\cdot) = V(\tau_n,\cdot) \to W_*(0,\cdot) \quad \text{strongly in } L^2(B_R).$$

If moreover the orbit has a uniform local L^6-bound (for example from a stronger local regularity hypothesis), then interpolation yields

$$V(\tau_n) \to V_* \quad \text{strongly in } L^3(B_R).$$

11.3.1. *Proof.* The compact embedding

$$H^1(B_R) \hookrightarrow\hookrightarrow L^2(B_R)$$

and continuous embedding

$$L^2(B_R) \hookrightarrow H^{-1}(B_R)$$

allow direct application of the Aubin–Lions–Simon theorem, giving

$$W_n \to W_* \quad \text{strongly in } L^2(0,1;L^2(B_R)).$$

The passage to traces at $s = 0$ requires the standard trace theory for functions in

$$L^2(0,1;H^1(B_R)) \cap H^1(0,1;H^{-1}(B_R)),$$

namely the Lions–Magenes result, which yields continuity into $L^2(B_R)$. This gives precompactness of $W_n(0,\cdot)$ in $L^2(B_R)$ after subsequence extraction.

Finally, if one also knows a uniform local L^6-bound, then the interpolation inequality

$$\|f\|_{L^3(B_R)} \le \|f\|_{L^2(B_R)}^{1/2} \|f\|_{L^6(B_R)}^{1/2}$$

upgrades strong L^2-convergence to strong L^3-convergence.
□

11.3.2. *Remark.* This lemma makes precise an important point: the **minimal** compactness closure needs only the $L^2_\tau H^1_y$ estimate, not an $L^\infty_\tau H^1_y$ estimate. However, to upgrade the convergence to L^3_{loc}, one still needs either:

- a uniform local L^6-bound, or
- some alternative compactness route in L^3_{loc}.

So the final theorem must be stated accordingly.

11.4. **Theorem A' (spacetime compactness formulation).** Let V, P solve the renormalized Navier–Stokes equation

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)\big(V + y\cdot\nabla V\big) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0$$

on $[\tau_0,\infty)\times\mathbb{R}^3$.

Fix $R_0 > 0$ and a cutoff $\phi_{R_0} \in C_c^\infty(\mathbb{R}^3)$, $\phi_{R_0} \ge 0$, $\phi_{R_0} \equiv 1$ on B_{R_0}, and define

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\,dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\,dy.$$

Assume:

11.4.1. *(H1') global L^3-normalization.*

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} = 1 \qquad \forall \tau \ge \tau_0.$$

11.4.2. *(H2') uniform L^3-tightness.* For every $\varepsilon > 0$, there exists $R_\varepsilon > 0$ such that

$$\sup_{\tau\ge\tau_0} \int_{|y|>R_\varepsilon} |V(y,\tau)|^3 < \varepsilon.$$

11.4.3. *(H3') local $L^2_\tau H^1_y$ control.* For every compact cylinder $Q_r \Subset Q_R \Subset \mathbb{R}^3\times(\tau_0,\infty)$, the renormalized Caccioppoli estimate yields

$$V \in L^2_{\mathrm{loc}}\big((\tau_0,\infty); H^1_{\mathrm{loc}}(\mathbb{R}^3)\big).$$

11.4.4. *(H4') local $L^2_\tau H^{-1}_y$ time-regularity.* For every $R > 0$,

$$\partial_\tau V \in L^2_{\mathrm{loc}}\big((\tau_0, \infty); H^{-1}(B_R)\big)$$

with uniform bounds on shifted windows.

11.4.5. *Derived pressure-tail control.* For every $R > 0$, Lemma 9.1 and the L^3-normalization give uniform control of the far-field pressure-tail term along the orbit.

11.4.6. *(H6') a local L^6-bound or equivalent L^3_{loc}-compactness upgrade.* For every fixed $R > 0$, either:

- $V(\tau_n)$ is uniformly bounded in $L^6(B_R)$ along every extracted sequence, or
- one has another mechanism yielding strong L^3_{loc}-compactness of the sampled orbit.

Then

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

11.4.7. *Proof.* Assume for contradiction that

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau < \infty.$$

By Lemma 6, there exists a sequence $\tau_n \to \infty$ such that

$$\tilde{\mathfrak{D}}_{R_0}(\tau_n) \to 0.$$

By Lemma 7,

$$\int_{B_{R_0}} |\nabla V(\tau_n)|^2 \to 0.$$

Now use (H3') and (H4') together with Lemma 13 to extract a subsequence such that

$$V(\tau_n) \to V_* \quad \text{strongly in } L^2_{\mathrm{loc}}(\mathbb{R}^3).$$

By (H6'), this convergence upgrades to

$$V(\tau_n) \to V_* \quad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3).$$

Now all hypotheses of Lemma 8 are satisfied:

- strong $L^2_{\mathrm{loc}} \cap L^3_{\mathrm{loc}}$ convergence,
- L^3-tightness by (H2'),
- normalization by (H1'),
- vanishing core dissipation from Lemma 7.

Lemma 8 yields a contradiction.
Therefore

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

□

11.5. **Auxiliary assumptions in the spacetime formulation.** With this formulation, the strongest unnecessary hypothesis has been removed.

The final theorem no longer requires the unproved orbit-level condition

$$\sup_{\tau} \|V(\tau)\|_{H^1(B_R)} < \infty.$$

What remains is the more natural and weaker package:

- $L^2_{\tau,\mathrm{loc}} H^1_{y,\mathrm{loc}}$ control from Caccioppoli,
- $L^2_{\tau,\mathrm{loc}} H^{-1}_{y,\mathrm{loc}}$ control of $\partial_\tau V$,
- and one mechanism upgrading local L^2-compactness to local L^3-compactness.

This is the correct tightened version of the exclusion theorem.

12. Scale-gauge consistency

12.1. **Canonical reference.** The scale-fixing condition used in the sequel is

$$G_{\mathrm{sc}}(V) := \int_{\mathbb{R}^3} |y|^{-p} |V(y)|^3 \, dy - \Theta_0 = 0, \qquad 0 < p < 3, \ \Theta_0 > 0.$$

Equivalently, in physical variables

$$\lambda(t)^p \int_{\mathbb{R}^3} |x - x_c(t)|^{-p} |u(x,t)|^3 \, dx = \Theta_0.$$

The corresponding modulation matrix is defined in G3, and its continuity is proved in G4.

12.2. **Central transversality identities.** The repaired-gauge transversality statements are collected as:

- $DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p \int |y|^{-p} |V|^3 \, dy$ (exact scale derivative),
- $DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0$ on the gauge surface,
- $M_{00}(V) = p\Theta_0$ and no annular lower bound is required for the scale row.

These are the only structural facts needed from the scale equation.

12.3. **External analytic inputs.** For a complete proof stack of these identities, including:

1. p-exact scaling under $V_\mu(y) = \mu V(\mu y)$,
2. Frechét differentiability of $V \mapsto \int |y|^{-p} |V|^3$,
3. exact formula $DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0$,
4. explicit modulation entries $M_{00}, M_{0\ell}, M_{j\ell}, M_{j0}$,
5. and the Schur-complement reduction from translation block + cross-terms,

see the repaired weighted-moment scale gauge supplement.

No additional gauge variant is used in the argument.

13. Good-window closure of the final compactness step

The endpoint trace compactness route can be replaced by a good-window selection route:

- finite total cost gives windows on which the average cost is small;
- local $L^2_\tau H^1_y$ control on the same windows gives good times with uniform local H^1;
- Rellich plus interpolation gives strong sampled-time L^3_{loc}-compactness.

This avoids the invalid inference from spacetime Aubin–Lions compactness to compactness at a fixed endpoint.

13.1. **Lemma 14 (finite cost gives vanishing-cost windows).** Let $D \in L^1([\tau_0, \infty))$ with $D \geq 0$. For every fixed window length $\ell > 0$, there exist times $s_n \to \infty$ such that

$$\int_{s_n}^{s_n+\ell} D(\tau)\, d\tau \to 0.$$

In particular, one may take $s_n = \tau_0 + n\ell$ after discarding finitely many initial windows.

13.1.1. *Proof.* Since $D \in L^1([\tau_0, \infty))$,

$$\int_T^\infty D(\tau)\, d\tau \to 0 \qquad \text{as } T \to \infty.$$

Therefore

$$\int_{\tau_0+n\ell}^{\tau_0+(n+1)\ell} D(\tau)\, d\tau \leq \int_{\tau_0+n\ell}^{\infty} D(\tau)\, d\tau \to 0.$$

□

13.2. **Lemma 15 (common good times on an exhaustion of balls).** Let $D \geq 0$ be a measurable cost density, and let

$$J_n = [s_n, s_n + \ell], \qquad \ell > 0, \qquad s_n \to \infty.$$

Assume

$$\int_{J_n} D(\tau)\, d\tau =: \delta_n \to 0.$$

Assume also that for every integer $m \geq 1$ there is $C_m < \infty$ such that

$$\sup_n \int_{J_n} \|V(\tau)\|^2_{H^1(B_m)}\, d\tau \leq C_m.$$

Then there exist times $\sigma_n \in J_n$ such that

$$D(\sigma_n) \to 0,$$

and for every fixed $m \geq 1$,

$$\sup_{n \geq m} \|V(\sigma_n)\|_{H^1(B_m)} < \infty.$$

Consequently, after passing to a diagonal subsequence, there is V_* such that

$$V(\sigma_n) \to V_* \quad \text{strongly in } L^2_{\mathrm{loc}}(\mathbb{R}^3)$$

and

$$V(\sigma_n) \to V_* \quad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3).$$

13.2.1. *Proof.* For each n, define

$$E_n^{\mathrm{cost}} := \{\tau \in J_n : D(\tau) \leq \delta_n^{1/2}\}.$$

By Chebyshev,

$$|J_n \setminus E_n^{\mathrm{cost}}| \leq \delta_n^{1/2}.$$

For each $m \leq n$, define

$$E_{n,m}^{H^1} := \left\{ \tau \in J_n : \|V(\tau)\|^2_{H^1(B_m)} \leq \frac{2^{m+2} C_m}{\ell} \right\}.$$

Again by Chebyshev,

$$|J_n \setminus E_{n,m}^{H^1}| \leq \frac{\ell}{2^{m+2}}.$$

Hence

$$\sum_{m=1}^{n} |J_n \setminus E_{n,m}^{H^1}| \le \frac{\ell}{4}.$$

For all sufficiently large n, $\delta_n^{1/2} < \ell/4$. Thus the intersection

$$E_n^{\mathrm{cost}} \cap \bigcap_{m=1}^{n} E_{n,m}^{H^1}$$

has positive measure. Choose σ_n in this intersection. Then

$$D(\sigma_n) \le \delta_n^{1/2} \to 0,$$

and for every fixed m, whenever $n \ge m$,

$$\|V(\sigma_n)\|_{H^1(B_m)}^2 \le \frac{2^{m+2} C_m}{\ell}.$$

For each fixed m, Rellich gives compactness of $V(\sigma_n)$ in $L^2(B_m)$. A diagonal subsequence yields strong convergence in L^2_{loc}.

The same uniform $H^1(B_m)$ bounds give uniform $L^6(B_m)$ bounds. Interpolation gives, for each fixed m,

$$\|V(\sigma_n) - V_*\|_{L^3(B_m)} \le C'_m \|V(\sigma_n) - V_*\|_{L^2(B_m)}^{1/2} \|V(\sigma_n) - V_*\|_{L^6(B_m)}^{1/2}.$$

The L^6 factor is uniformly bounded and the L^2 factor tends to zero, so the convergence is strong in $L^3(B_m)$. Since m was arbitrary, the convergence is strong in L^3_{loc}.

□

13.3. **Theorem A" (good-window compact Type II criterion).** Let V, P solve the renormalized Navier–Stokes equation

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0$$

on $[\tau_0, \infty) \times \mathbb{R}^3$.

Fix $R_0 > 0$ and define

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\, dy.$$

Assume:

13.3.1. *(GW1) global L^3-normalization.*

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} = 1 \qquad \forall \tau \ge \tau_0.$$

13.3.2. *(GW2) uniform L^3-tightness.* For every $\varepsilon > 0$, there exists $R_\varepsilon > 0$ such that

$$\sup_{\tau \ge \tau_0} \int_{|y| > R_\varepsilon} |V(y,\tau)|^3\, dy < \varepsilon.$$

13.3.3. *(GW3) uniform local $L^2_\tau H^1_y$ bounds on unit windows.* For every integer $m \ge 1$, there exists $C_m < \infty$ such that

$$\sup_{n \ge 0} \int_{\tau_0 + n}^{\tau_0 + n + 1} \|V(\tau)\|_{H^1(B_m)}^2\, d\tau \le C_m.$$

Then

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

13.3.4. *Proof.* Assume for contradiction that

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau < \infty.$$

By Lemma 14 with $D = \tilde{\mathfrak{D}}_{R_0}$ and $\ell = 1$,

$$\delta_n := \int_{\tau_0+n}^{\tau_0+n+1} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau \to 0.$$

Apply Lemma 15 to the windows

$$J_n = [\tau_0 + n, \tau_0 + n + 1].$$

There exist $\sigma_n \in J_n$ such that

$$\tilde{\mathfrak{D}}_{R_0}(\sigma_n) \to 0,$$

and, after passing to a subsequence,

$$V(\sigma_n) \to V_* \quad \text{strongly in } L^2_{\mathrm{loc}}(\mathbb{R}^3) \cap L^3_{\mathrm{loc}}(\mathbb{R}^3).$$

Because $\phi_{R_0} \equiv 1$ on B_{R_0},

$$\nu \int_{B_{R_0}} |\nabla V(\sigma_n)|^2\, dy \le \tilde{\mathfrak{D}}_{R_0}(\sigma_n) \to 0.$$

Thus the low-dissipation rigidity theorem, Lemma 8, applies to the sequence $V(\sigma_n)$: the sequence is globally L^3-normalized by (GW1), uniformly L^3-tight by (GW2), strongly compact locally in $L^2 \cap L^3$, and has vanishing core dissipation on B_{R_0}. Lemma 8 gives a contradiction.

Therefore

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

□

13.3.5. *Consequence.* Theorem A" removes the separate sampled-time compactness and H^{-1}-time-regularity assumptions from the final contradiction step. Those estimates remain useful for other compactness routes, but the good-window proof needs only normalization, tightness, and uniform local $L^2_\tau H^1_y$ bounds on the same windows as the vanishing average cost.

13.4. **Corollary 16 (contrapositive classification of finite-cost Type II candidates).** Assume a normalized renormalized Type II candidate has finite total renormalization cost:

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau < \infty.$$

Assume also

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} = 1 \qquad \forall \tau \ge \tau_0.$$

Then at least one of the following must occur.

13.4.1. *(C1) failure of global critical tightness.* There exists $\varepsilon_0 > 0$ such that for every $R > 0$ there is a time $\tau_R \ge \tau_0$ with

$$\int_{|y|>R} |V(y, \tau_R)|^3\, dy \ge \varepsilon_0.$$

13.4.2. *(C2) failure of local windowed H^1 control.* There exists an integer $m \geq 1$ such that

$$\sup_{n\geq 0} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|_{H^1(B_m)}^2 \, d\tau = \infty.$$

Equivalently, every finite-cost normalized Type II candidate is either noncompact/radiative in the critical L^3 topology, or locally rough in the renormalized core.

13.4.3. *Proof.* If neither (C1) nor (C2) holds, then V is uniformly L^3-tight and satisfies the uniform local $L^2_\tau H^1_y$ window bounds of (GW3). Together with the normalization hypothesis, all assumptions of Theorem A" hold. Theorem A" gives

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) \, d\tau = \infty,$$

contradicting the finite-cost assumption. Therefore at least one of (C1) or (C2) must occur. □

13.4.4. *Interpretation.* Corollary 16 narrows the remaining finite-cost Type II scenarios to two structural families:

(1) **Noncompact/radiative Type II:** critical L^3 mass escapes every bounded renormalized region. This includes radiation tails, outward-moving secondary profiles, and no-splitting failures.
(2) **Rough-core Type II:** the orbit may remain L^3-tight, but local windowed H^1 control fails on some bounded renormalized core.

Failure of H^{-1}-time regularity alone is not an admissible survivor mechanism for the good-window closure. It matters only insofar as it reflects failure to obtain the local windowed H^1 bounds or other hypotheses needed upstream.

Appendix A. Repaired weighted-moment scale gauge

We record the scale-gauge layer used in the compact Type II analysis with fully explicit assumptions and proof steps.

A.1. **Setup and admissibility class.** Fix

$$0 < p < 3, \qquad \Theta_0 > 0, \qquad w(y) := |y|^{-p}.$$

For a vector-valued profile $V : \mathbb{R}^3 \to \mathbb{R}^m$ ($m \geq 1$), define

$$G_{\mathrm{sc}}(V) := \int_{\mathbb{R}^3} w(y)|V(y)|^3 \, dy - \Theta_0.$$

Set

$$I_p(V) := \int_{\mathbb{R}^3} |y|^{-p}|V(y)|^3 \, dy.$$

Throughout this appendix, all integrals are over $\mathbb{R}^3$ and all formulas are written for real-valued profiles. For complex-valued V, replace $V \cdot W$ by $\Re(V \cdot \overline{W})$ everywhere.

The class $\mathcal{A}_p$ used in the gauge formulas below is

$$\mathcal{A}_p := \left\{ V : I_p(V) < \infty, \ \int_{\mathbb{R}^3} |y|^{-p}|V| \, |y \cdot \nabla V| \, dy < \infty \right\}.$$

This class contains all smooth compactly supported profiles and is the natural admissibility class for the repaired gauge in the present analysis.

Define also the scaling generator

$$Z_{\mathrm{sc}}(V) := V + y \cdot \nabla V.$$

A.2. **Theorem R1 (exact scaling law).** For $\mu > 0$, set $V_\mu(y) := \mu\, V(\mu y)$. If $V \in \mathcal{A}_p$, then

$$I_p(V_\mu) = \mu^p I_p(V).$$

Consequently,

$$G_{\rm sc}(V_\mu) = \mu^p \int_{\mathbb{R}^3} |y|^{-p}|V(y)|^3\, dy - \Theta_0.$$

A.2.1. *Proof.*

$$I_p(V_\mu) = \int_{\mathbb{R}^3} |y|^{-p}|\mu V(\mu y)|^3\, dy = \mu^3 \int_{\mathbb{R}^3} |y|^{-p}|V(\mu y)|^3\, dy.$$

With $z = \mu y$, $dy = \mu^{-3}dz$, and $|y|^{-p} = \mu^p|z|^{-p}$, hence

$$I_p(V_\mu) = \mu^3 \int_{\mathbb{R}^3} \mu^{-p}|z|^{-p}|V(z)|^3\, \mu^{-3}dz = \mu^p I_p(V).$$

Substituting into $G_{\rm sc}$ gives the second identity. □

A.3. **Theorem R2 (Fréchet derivative of $G_{\rm sc}$).** Let $V, W \in \mathcal{A}_p$ and assume $W \in L^3(\mathbb{R}^3, w\, dy)$. Then the directional derivative exists:

$$DG_{\rm sc}(V)[W] = 3 \int_{\mathbb{R}^3} |y|^{-p}|V(y)|\, V(y) \cdot W(y)\, dy.$$

A.3.1. *Proof.* For $h \neq 0$,

$$\frac{G_{\rm sc}(V + hW) - G_{\rm sc}(V)}{h} = \int_{\mathbb{R}^3} |y|^{-p} \frac{|V + hW|^3 - |V|^3}{h}\, dy.$$

For each fixed y, by the one-dimensional mean-value formula along $t \mapsto |V + tW|^3$,

$$\frac{|V + hW|^3 - |V|^3}{h} = 3\int_0^1 |V + \theta hW|\, (V + \theta hW) \cdot W\, d\theta.$$

Thus

$$\left|\frac{|V + hW|^3 - |V|^3}{h}\right| \le 3(|V| + |h||W|)^2|W| \le 6\big(|V|^2|W| + |W|^3\big).$$

Now $|V|^2|W| \le \frac{2}{3}|V|^3 + \frac{1}{3}|W|^3$, so

$$6\big(|V|^2|W| + |W|^3\big) \le 6|V|^3 + 10|W|^3.$$

Multiplying by w, the dominating integrand is integrable by the class assumptions. Hence dominated convergence applies and yields

$$\lim_{h\to 0} \frac{G_{\rm sc}(V + hW) - G_{\rm sc}(V)}{h} = \int w\, \frac{d}{dh}\bigg|_{h=0} |V + hW|^3\, dy = 3\int w|V|\, V \cdot W\, dy.$$

That is exactly the stated formula. □

A.4. **Lemma R2.5 (equivalent scaling derivative identity for $Z_{\rm sc}$).** For $V \in \mathcal{A}_p$, the map $\mu \mapsto V_\mu$ satisfies

$$\frac{d}{d\mu}\bigg|_{\mu=1} V_\mu = V + y \cdot \nabla V = Z_{\rm sc}(V).$$

Hence

$$\frac{d}{d\mu}\bigg|_{\mu=1} G_{\rm sc}(V_\mu) = DG_{\rm sc}(V)\big[Z_{\rm sc}(V)\big].$$

A.4.1. *Proof.* Direct differentiation in the scaling formula $V_\mu(y) = \mu V(\mu y)$ gives

$$\frac{d}{d\mu} V_\mu(y) = V(\mu y) + \mu\, y \cdot \nabla V(\mu y),$$

so at $\mu = 1$:

$$\frac{d}{d\mu} V_\mu(y)\Big|_{\mu=1} = V(y) + y \cdot \nabla V(y) = Z_{\mathrm{sc}}(V)(y).$$

Applying the chain rule for directional derivatives of G_{sc} in class $\mathcal{A}_p$ gives the second identity. □

A.5. **Theorem R3 (exact scale transversality).** Assume $V \in \mathcal{A}_p$ is smooth and compactly supported. Then

$$DG_{\mathrm{sc}}(V)\big[Z_{\mathrm{sc}}(V)\big] = p \int_{\mathbb{R}^3} |y|^{-p} |V(y)|^3 \, dy.$$

If $G_{\mathrm{sc}}(V) = 0$, then

$$DG_{\mathrm{sc}}(V)\big[Z_{\mathrm{sc}}(V)\big] = p\Theta_0.$$

A.5.1. *Proof.* Apply Theorem R2 with $W = Z_{\mathrm{sc}}(V)$:

$$DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = 3 \int w |V|\, V \cdot (V + y \cdot \nabla V)\, dy.$$

Use $3|V|\, V \cdot (V + y \cdot \nabla V) = 3|V|^3 + y \cdot \nabla(|V|^3)$:

$$DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = 3 \int w |V|^3 \, dy + \int w\, y \cdot \nabla(|V|^3)\, dy.$$

Integration by parts is valid for compactly supported V:

$$\int w\, y \cdot \nabla(|V|^3)\, dy = - \int |V|^3\, \nabla \cdot (wy)\, dy.$$

Now

$$\nabla \cdot (wy) = 3|y|^{-p} + y \cdot \nabla(|y|^{-p}) = 3|y|^{-p} - p|y|^{-p} = (3-p)|y|^{-p},$$

so

$$DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = 3 \int w|V|^3\, dy - (3-p) \int w|V|^3\, dy = p \int w|V|^3\, dy.$$

Imposing $G_{\mathrm{sc}}(V) = 0$ yields $\int w|V|^3 = \Theta_0$, hence the final identity. □

By density of compactly supported smooth profiles in $\mathcal{A}_p$, this identity extends to general admissible V under the same integrability class.

A.6. **Theorem R4 (gauge-transversality constant).** On the gauge surface $G_{\mathrm{sc}}(V) = 0$,

$$DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0 =: c_0 > 0.$$

A.6.1. *Corollary.* In particular,

$$DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] \geq c_0.$$

A.7. **Theorem R5 (modulation entries with the repaired scale gauge).** Let $T_\ell(V) := \partial_{y_\ell} V$. Under $V \in \mathcal{A}_p \cap W^{1,3}_{\mathrm{loc}}$ with sufficient decay to justify the integration steps (in particular, those steps in Theorem R3 and in the compact Type II analysis), define

$$M_{00}(V) := DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)], \qquad M_{0\ell}(V) := DG_{\mathrm{sc}}(V)[T_\ell(V)], \ \ell = 1,2,3.$$

Then

$$M_{00}(V) = p \int_{\mathbb{R}^3} |y|^{-p} |V|^3 \, dy = p\Theta_0,$$

and

$$M_{0\ell}(V) = 3 \int_{\mathbb{R}^3} |y|^{-p} |V| \, V \cdot \partial_\ell V \, dy = \int_{\mathbb{R}^3} |y|^{-p} \partial_\ell(|V|^3) \, dy = p \int_{\mathbb{R}^3} y_\ell |y|^{-p-2} |V(y)|^3 \, dy.$$

A.7.1. *Proof.* The formula for M_{00} is Theorem R3. For $M_{0\ell}$, first apply Theorem R2:

$$M_{0\ell} = 3 \int |y|^{-p} |V| \, V \cdot \partial_\ell V \, dy = \int |y|^{-p} \partial_\ell(|V|^3) \, dy.$$

An integration by parts identity with compact support in the $\mathcal{A}_p$ approximation gives

$$\int |y|^{-p} \partial_\ell(|V|^3) \, dy = -\int |V|^3 \, \partial_\ell(|y|^{-p}) \, dy = p \int y_\ell |y|^{-p-2} |V|^3 \, dy.$$

Hence the stated formula. □

A.8. **Theorem R6 (continuity of repaired scale-functionals).** Assume

$$V_n \to V \text{ in } L^3(\mathbb{R}^3, |y|^{-p} dy), \qquad |y|^{-p-2} y \, |V_n|^3 \to |y|^{-p-2} y \, |V|^3 \text{ in } L^1_{\mathrm{loc}},$$

and that $V_n \to V$ in the local $W^{1,3}$-topology used for the translation block (or the continuity hypotheses from Parts VIII–G4). Then each repaired scale entry satisfies

$$M_{00}(V_n) \to M_{00}(V), \qquad M_{0\ell}(V_n) \to M_{0\ell}(V),$$

and the full modulation matrix built from $G_{\mathrm{sc}}, G_1, G_2, G_3$ is continuous in n.

A.8.1. *Proof.*

- $M_{00}(V_n)$ depends linearly on $I_p(V_n)$, hence continuity by weighted L^1 convergence.
- $M_{0\ell}(V_n)$ depends on the weighted first moments $\int y_\ell |y|^{-p-2} |V_n|^3$, which are continuous by the second assumption.
- translation block entries are controlled exactly as in the existing local continuity argument (same assumptions as in Parts VIII–G4). Combining these yields continuity of every entry and therefore of the full matrix. □

A.9. **Theorem R7 (full gauge matrix nondegeneracy reduction).** Define

$$M(V) = \begin{pmatrix} M_{00}(V) & M_{01}(V) & M_{02}(V) & M_{03}(V) \\ M_{10}(V) & M_{11}(V) & M_{12}(V) & M_{13}(V) \\ M_{20}(V) & M_{21}(V) & M_{22}(V) & M_{23}(V) \\ M_{30}(V) & M_{31}(V) & M_{32}(V) & M_{33}(V) \end{pmatrix}$$

using the repaired gauge and the standard centering gauge. Assume along the orbit:

(1) $M_{00}(V) = p\Theta_0 \geq c_0 > 0$.
(2) translation block $A(V) := (M_{j\ell}(V))_{1 \leq j,\ell \leq 3}$ is invertible with
$$\|A(V)^{-1}\| \leq C_A.$$
(1) $S(V) := M_{00}(V) - u(V)^T A(V)^{-1} v(V) \geq \sigma_0 > 0$, where $u = (M_{01}, M_{02}, M_{03})^T$, $v = (M_{10}, M_{20}, M_{30})$.

Then $M(V)$ is uniformly invertible and the usual Schur complement estimate applies.

A.9.1. *Proof.* Write $M = \left(\begin{smallmatrix} \alpha & u^T \\ v & A \end{smallmatrix}\right)$ with $\alpha = M_{00}$. The assumption A^{-1} exists gives

$$M^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}u^T A^{-1} \\ -A^{-1}vS^{-1} & A^{-1} + A^{-1}vS^{-1}u^T A^{-1} \end{pmatrix}, \qquad S = \alpha - u^T A^{-1} v.$$

Uniform lower bounds on α, S and upper bounds on $\|A^{-1}\|$, $\|u\|$, $\|v\|$ give a uniform bound on $\|M^{-1}\|$. □

A.10. **Use in the compact Type II analysis.** In the compactness argument, the repaired gauge is used exclusively through:

(1) $G_{\mathrm{sc}}(V) = 0$,
(2) The formula $DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0$,
(3) The explicit scale/translation matrix entries of Theorem R5,
(4) Schur-complement reduction of Theorem R7.

No other scale-gauge variant is introduced.

Appendix B. Strong local L^3-compactness upgrade

This appendix records the sampled-sequence upgrade from local compactness in L^2 to compactness in L^3.

Assumptions and notation are aligned with the compactness criterion.

Let $R > 0$, $I = (0, 1)$, and for each index n define shifted trajectories

$$W_n(s, y) := V(\tau_n + s, y), \qquad (s, y) \in I \times B_{2R},$$

where $\tau_n \to \infty$.

B.1. **T7.1 (spacetime compactness in $L^2_t L^2$).** Assume:

(1) $\sup_n \|W_n\|_{L^2(I;H^1(B_{2R}))} < \infty$.
(2) $\sup_n \|\partial_s W_n\|_{L^2(I;H^{-1}(B_R))} < \infty$.

Then there exists a subsequence (not relabelled) and a limit $W \in L^2(I; L^2(B_R))$ such that

$$W_n \to W \quad \text{strongly in } L^2(I; L^2(B_R)).$$

B.1.1. *Proof.* Compactness of embeddings $H^1(B_R) \hookrightarrow\hookrightarrow L^2(B_R) \hookrightarrow H^{-1}(B_R)$ with bounds in (1)–(2) gives this from Aubin–Lions–Simon, as in Lemma 11 and Lemma 13.

B.2. **T7.2 (spacetime upgrade to $L^2_t L^3$ from local L^6-type control).** Assume T7.1 hypotheses and additionally

(1) $\sup_n \|W_n\|_{L^2(I;L^6(B_{2R}))} < \infty$.

Then, passing to the same subsequence,

$$W_n \to W \quad \text{strongly in } L^2(I; L^3(B_R)).$$

B.2.1. *Proof.* By assumption 3 and reflexivity, after passing to a further subsequence

$$W_n \rightharpoonup \widetilde{W} \quad \text{weakly in } L^2(I; L^6(B_R)).$$

The strong $L^2(I; L^2(B_R))$-limit from T7.1 identifies $\widetilde{W} = W$. Hence

$$W \in L^2(I; L^6(B_R))$$

and $\|W_n - W\|_{L^2(I;L^6(B_R))}$ is uniformly bounded.

By Gagliardo–Nirenberg on bounded sets,

$$\|W_n - W\|_{L^3(B_R)} \le C(R) \|W_n - W\|^{1/2}_{L^2(B_R)} \|W_n - W\|^{1/2}_{L^6(B_R)}.$$

Square and integrate in s:

$$\int_0^1 \|W_n - W\|_{L^3}^2 ds \le C \int_0^1 \|W_n - W\|_{L^2} \|W_n - W\|_{L^6}\, ds.$$

Apply Hölder in time to obtain

$$\|W_n - W\|_{L^2(I;L^3(B_R))}^2 \le C \|W_n - W\|_{L^2(I;L^2(B_R))} \|W_n - W\|_{L^2(I;L^6(B_R))}.$$

The first factor tends to zero by T7.1 and the second is uniformly bounded, so the left side tends to zero.

B.3. **T7.3 (sampled-time L^3 upgrade from sampled-time L^2 compactness).** Assume that, after passing to a subsequence, the sampled states satisfy

(1) $W_n(0,\cdot) \to f$ strongly in $L^2(B_R)$.
(2) $\sup_n \|W_n(0,\cdot)\|_{L^6(B_R)} < \infty$.

Then

$$W_n(0,\cdot) \to f \quad \text{strongly in } L^3(B_R).$$

B.3.1. *Proof.* By assumption 5 and weak compactness, the strong L^2-limit f belongs to $L^6(B_R)$, and

$$\sup_n \|W_n(0,\cdot) - f\|_{L^6(B_R)} < \infty.$$

Interpolating at $s = 0$,

$$\|W_n(0) - f\|_{L^3(B_R)} \le C(R) \|W_n(0) - f\|_{L^2(B_R)}^{1/2} \|W_n(0) - f\|_{L^6(B_R)}^{1/2}.$$

The second factor is bounded and the first tends to zero by assumption 4, so the right-hand side tends to zero.

B.3.2. *Endpoint limitation.* T7.1 and T7.2 are spacetime compactness statements. They do not, by themselves, imply convergence at the fixed time $s = 0$. A separate sampled-time L^2-compactness mechanism is required for T7.3.

A sufficient mechanism is Simon compactness in the form:

(1) $\sup_n \|W_n\|_{L^\infty(I;H^1(B_R))} < \infty$.
(2) $\sup_n \|\partial_s W_n\|_{L^q(I;H^{-1}(B_R))} < \infty$ for some $q > 1$.

Then (W_n) is precompact in $C([0,1]; L^2(B_R))$, so assumption 4 follows after subsequence extraction. Without such an additional input, fixed-time convergence at $s = 0$ is not a consequence of Aubin–Lions in $L^2(I; L^2)$.

B.4. **T7.4 (good-time replacement for sampled-time compactness).** Let $J_n = [s_n, s_n + \ell]$ be time windows with fixed length $\ell > 0$, and let $\delta_n \downarrow 0$. Assume for a fixed radius R:

(1) The localized cost is small on average:

$$\int_{J_n} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau \le \delta_n.$$

(1) The local H^1 energy is uniformly integrable on the same windows:

$$\int_{J_n} \|V(\tau)\|_{H^1(B_R)}^2\, d\tau \le C_R.$$

Then there are times $\sigma_n \in J_n$ such that

$$\tilde{\mathfrak{D}}_{R_0}(\sigma_n) \to 0, \qquad \|V(\sigma_n)\|_{H^1(B_R)} \le \left(\frac{2C_R}{\ell}\right)^{1/2}.$$

Consequently, after passing to a subsequence,

$$V(\sigma_n) \to V_* \quad \text{strongly in } L^2(B_R)$$

and

$$V(\sigma_n) \to V_* \quad \text{strongly in } L^3(B_R).$$

B.4.1. *Proof.* Define

$$E_n := \left\{ \tau \in J_n : \tilde{\mathfrak{D}}_{R_0}(\tau) \le \delta_n^{1/2} \right\}.$$

By Chebyshev,

$$|J_n \setminus E_n| \le \delta_n^{1/2}.$$

For all sufficiently large n, $|E_n| \ge \ell/2$.

Since

$$\int_{E_n} \|V(\tau)\|_{H^1(B_R)}^2 \, d\tau \le C_R,$$

there exists $\sigma_n \in E_n$ such that

$$\|V(\sigma_n)\|_{H^1(B_R)}^2 \le \frac{2C_R}{\ell}.$$

Because $\sigma_n \in E_n$,

$$\tilde{\mathfrak{D}}_{R_0}(\sigma_n) \le \delta_n^{1/2} \to 0.$$

The uniform $H^1(B_R)$ bound gives, by Rellich, a subsequence converging strongly in $L^2(B_R)$.

The same $H^1(B_R)$ bound gives a uniform $L^6(B_R)$ bound. Interpolation gives

$$\|V(\sigma_n) - V_*\|_{L^3(B_R)} \le C_R \|V(\sigma_n) - V_*\|_{L^2(B_R)}^{1/2} \|V(\sigma_n) - V_*\|_{L^6(B_R)}^{1/2}.$$

The L^6 factor is uniformly bounded and the L^2 factor tends to zero, proving strong $L^3(B_R)$ convergence.

B.4.2. *Role in the compactness proof.* T7.4 avoids endpoint trace compactness. Instead of first fixing times τ_n and then trying to prove compactness of $V(\tau_n)$, one selects low-cost times inside windows where the same windows also carry local H^1 control. This discharges the abstract sampled-time L^2-compactness hypothesis in T7.3 whenever the proof can arrange common good windows.

B.5. **Corollary for renormalized times.** Assume that for each fixed local radius R, the shifted sequence satisfies T7.1 and the sampled states satisfy the two assumptions of T7.3. Then, after passing to a subsequence,

$$V(\tau_n) \to V_* \quad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3)$$

where V_* is the local L^2-limit of the sampled states.

For clarity, the chain is:

(1) Caccioppoli-type input gives (1).
(2) Time-regularity input (T6) gives (2).
(3) A local L^6-type upgrade gives (3) and sampled-time L^6 boundedness.
(4) T7.1 and T7.2 give spacetime L^3-compactness.
(5) An additional sampled-time L^2-compactness mechanism gives T7.3.
(6) T7.3 gives sampled-time strong L^3-compactness needed for the low-dissipation contradiction theorem.

This gives the local compactness upgrade used in the low-dissipation contradiction.

B.6. **Corollary using good times.** If there is a single sequence of selected times σ_n such that, for every fixed local radius R,

$$\sup_n \|V(\sigma_n)\|_{H^1(B_R)} < \infty$$

and

$$\tilde{\mathfrak{D}}_{R_0}(\sigma_n) \to 0,$$

then

$$V(\sigma_n) \to V_* \quad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3)$$

after diagonal subsequence extraction. T7.4 supplies such a sequence for any fixed radius; to get full L^3_{loc} compactness, the good-time selection must be made on common windows with local H^1 control on an exhaustion of balls. In that case, T7.4 replaces the separate sampled-time compactness input in T7.3.

References